POWER MODULE

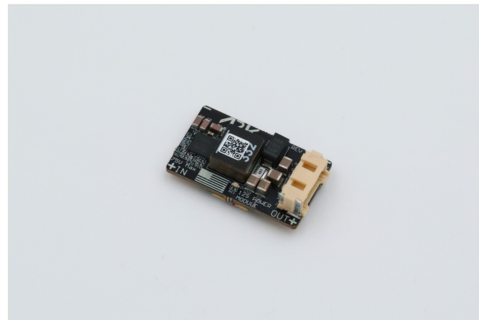
ARK 12S PAB Power Module

5.2 V / 6 A regulated supply and digital power monitor for the Pixhawk Autopilot Bus

EXPOSED INTERFACES BATTERY I2C 5 V

The USA-built ARK 12S PAB Power Module delivers regulated 5.2 V power and high-accuracy digital battery monitoring to any Pixhawk Autopilot Bus carrier. A high-current battery path rated 100 A continuous passes through a 0.1 mΩ shunt monitored by a TI INA238, while a synchronous buck regulator provides 6 A of clean 5.2 V for the avionics.

The 6-pin Molex CLIK-Mate output matches the ARK PAB Carrier power pinout exactly, so the module drops in without adapters. Parallel modules increase continuous battery current beyond 100 A.



01 KEY FEATURES

- 100 A continuous battery current path (parallel for higher)
- TI LM70880 synchronous buck — 5.2 V / 6 A regulated output
- TI INA238 digital power monitor with 0.1 mΩ shunt (I²C)
- 10 V – 75 V input range (12S capable)
- 6-pin Molex CLIK-Mate output matching PAB pinout
- Solder-pad battery input and output
- Includes 6-pin Molex cable

02 SPECIFICATIONS

ELECTRICAL	
Input voltage	10 V – 75 V
Battery current	100 A continuous (20 °C ambient)
Regulated output	5.2 V @ 6 A continuous
Power monitor	INA238, 0.1 mΩ shunt, I²C (default 0x45)
Operating temperature	−40 °C to +85 °C

MECHANICAL	
Dimensions	37 × 22 × 13 mm
Weight	11.5 g

AT A GLANCE	
Input	10 – 75 V
Battery current	100 A cont.
Output	5.2 V / 6 A
Monitor	INA238 0.1 mΩ
Connector	6-pin Molex
Dimensions	37 × 22 × 13 mm
Weight	11.5 g
Hardware rev	Rev 3.1

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-12s-pab-power-module/

DOCS docs.arkelectron.com/products/power/ark-12s-pab-power-module

CAD github.com/ARK-Electronics/ark-hardware

03 OUTPUT CONNECTOR PINOUT

PIN	SIGNAL	VOLTAGE	NOTES
1	VBRICK	5.2 V	Regulated output
2	VBRICK	5.2 V	Regulated output
3	SCL	3.3 V	I²C clock to INA238
4	SDA	3.3 V	I²C data to INA238
5	GND	GND	
6	GND	GND	

Connector: Molex 5024940670 6-position CLIK-Mate. Matches ARK PAB Carrier power pinout.

04 NOTES

- Active cooling is recommended when operating at high battery current or high 5 V regulator load.
- For continuous battery current above 100 A, operate multiple modules in parallel.
- When using 12S batteries, solder the included capacitor and TVS diode to the battery-output pads (observe polarity).
- Battery input and output are solder pads for high-current connections.

05 ALTERNATIVES

PRODUCT	KEY DIFFERENCE	WHEN TO CHOOSE
ARK PAB Power Module	Lower voltage / 60 A path	6S systems, lower current
ARK 12S Payload Power Module	Adds 12 V / 6 A payload rail	Need both avionics 5 V and payload 12 V

ARK-DS-ARK-12S-PAB-Power-Module | Issue 2026-08 | Compiled from the ARK 12S PAB Power Module Production schematic Rev 3.1 (24 Apr 2026), the Production bill of materials, the production STEP model, and ARK Electronics published documentation. Specifications are subject to change without notice; confirm against current published documentation before design commitment. ARK Electronics products are designed and manufactured in the USA. © 2026 ARK Electronics LLC.